Monish Sinha

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Education

Case Western Reserve University Aug 2023 – May 2027
B.S. Data Science, Math minor | GPA: 3.5/4.0 *Cleveland, OH*

University College Dublin Jan 2026 – May 2026
Study Abroad | Big Data, Information Visualization *Dublin, IE*

Relevant Experience

PNC Financial Services June 2026 – Present
Full-Stack Software Engineer *Pittsburgh, PA*

- Delivered over **$2M** in vendor cost savings by developing a Java 21 code conversion engine that migrated legacy Hive workloads to Iceberg-compatible Spark SQL using Azure AI Foundry.
- Reduced CR report preparation time by over **90%** (from 1+ hour to under 5 minutes) by designing an LLM-powered workflow integrating Jira, Bitbucket, and Azure AI Foundry.
- Cut Tableau dashboard archival time by **90%** (from 20 minutes to 2 minutes) by designing an autonomous data pipeline that archived dashboards to Bitbucket before expiration while preserving version-controlled backups.
- Accelerated enterprise AI and automation development by engineering reusable Java 21 libraries for Jira, Bitbucket, and GitHub Enterprise REST APIs.

Sherwin-Williams May 2025 – Aug 2025
AI Governance Intern *Cleveland, OH*

- Reduced AI proposal evaluation time by **50%** by building a predictive GenAI cost-modeling application using Copilot and Azure AI.
- Reduced unnecessary GitHub Copilot licenses by **15%** by developing Power BI dashboards that tracked Microsoft 365 and Copilot adoption.
- Improved AI cost-reporting accuracy by implementing an Azure AI tagging framework that automatically attributed cloud costs to specific AI services.
- Strengthened responsible AI practices for **64,000+** employees by coauthoring AI governance guidelines adopted across the organization.

Technical Projects

Limit Order Book Matching Engine | *Python, pytest, GitHub Actions* Fall 2026 – Present

- Leading a 4-person CWRU Quants team building a price-time priority matching engine; designed the core interfaces and roadmap.
- Own architecture and code review, enforcing interface contracts and CI quality gates across all team contributions.

U.S. Voting Patterns Analysis | *Python, Spark* Spring 2026

- Built Spark MapReduce pipelines across 47,000+ county-year observations, 500+ variables, and 3,143 counties, achieving up to 3.5× faster processing than a single-node implementation.

Additional Experience

Conduct Review Board, Board Member Nov 2024 – Present

- Reviewed academic and conduct cases with faculty to uphold university standards.

CWRU Quants, Executive Member Nov 2024 – Present

- Developed an event-driven Python backtesting framework for quantitative trading strategy research.

Skills & Interests

Languages: Python, SQL, Java, TypeScript, JavaScript
AI & Agents: Azure AI Foundry, Model Context Protocol, LLM Applications, Microsoft Copilot Studio
Data: Apache Spark, Spark SQL, Apache Iceberg, Apache Hive, Oracle Database
Frameworks & Tools: Spring Boot, Next.js, Node.js, Git, Maven, Linux, Jira & Bitbucket APIs
Interests: Rock climbing, backpacking, photography, tennis